

Project1_Academic Performance Analysis

October 3, 2026

1 Project 1

1.1 Project Overview

In this project, you are helping me by analyzing student academic performance using basic Python programming concepts. We work with a 2D list dataset, modify data, compute statistics, and analyze how adding and removing students affects class averages.

1.1.1 Dataset Description

- There are 20 students in this class.
- Each row represents one student:[Student Name, Math Score, Science Score, English Score]
- They are in alphabetical order.

1.1.2 Step 1: Create the Dataset

```
[1]: students = [  
    ["Alice", 85, 90, 88],  
    ["Bob", 78, 82, 80],  
    ["Charlie", 92, 88, 91],  
    ["Diana", 88, 91, 87],  
    ["Ethan", 70, 75, 72],  
    ["Fiona", 95, 94, 96],  
    ["George", 60, 65, 63],  
    ["Hannah", 83, 85, 84],  
    ["Ian", 76, 79, 78],  
    ["Julia", 89, 90, 92],  
    ["Kevin", 82, 80, 81],  
    ["Liam", 90, 92, 91],  
    ["Mia", 75, 78, 76],  
    ["Noah", 68, 70, 72],  
    ["Olivia", 91, 89, 93],  
    ["Paul", 73, 74, 75],  
    ["Quinn", 87, 86, 88],  
    ["Ryan", 79, 81, 80],  
    ["Sophia", 94, 95, 93],  
    ["Tyler", 66, 69, 67]  
]
```

1.1.3 Step 2: Correct Incorrect Data Using Indexing

We found that **George's English score** is wrong! Let's replace it with **68**.

1.1.4 Step 3: Calculate Individual Total and Average Scores

Let's check each student's average score! Use a loop and arithmetic operations to calculate **each student's total score** and **average score**. Print each student's name with his/her total score and average score.

1.1.5 Step 4: Assign Letter Grades

Each student needs to know his/her letter grade.

Use `if / elif / else` to classify students. Recalculate each average score over three subjects, and **assign grades** based on:

Average	Grade
90+	A
80-89	B
70-79	C
Below 70	F

Print each student's name and letter grade.

```
[2]: #step 2
students[6][3] = 68
#step 3
for student in students:
    total = sum(student[1:4])
    average = total / 3
    print(student[0], "Total:", total, "Average:", round(average,2), sep="\t")
#step 4
print("-----")
for index in range(0,len(students)):
    total = sum(students[index][1:4])
    average = total / 3
    if average < 70:
        grade = "F"
    elif average < 80:
        grade = "C"
    elif average < 90:
        grade = "B"
    else:
        grade = "A"
    students[index].append(grade)
    print(students[index][0], grade, sep="\t")
```

Alice	Total:	263	Average:	87.67
Bob	Total:	240	Average:	80.0
Charlie	Total:	271	Average:	90.33
Diana	Total:	266	Average:	88.67
Ethan	Total:	217	Average:	72.33
Fiona	Total:	285	Average:	95.0
George	Total:	193	Average:	64.33
Hannah	Total:	252	Average:	84.0
Ian	Total:	233	Average:	77.67
Julia	Total:	271	Average:	90.33
Kevin	Total:	243	Average:	81.0
Liam	Total:	273	Average:	91.0
Mia	Total:	229	Average:	76.33
Noah	Total:	210	Average:	70.0
Olivia	Total:	273	Average:	91.0
Paul	Total:	222	Average:	74.0
Quinn	Total:	261	Average:	87.0
Ryan	Total:	240	Average:	80.0
Sophia	Total:	282	Average:	94.0
Tyler	Total:	202	Average:	67.33

Alice	B
Bob	B
Charlie	A
Diana	B
Ethan	C
Fiona	A
George	F
Hannah	B
Ian	C
Julia	A
Kevin	B
Liam	A
Mia	C
Noah	C
Olivia	A
Paul	C
Quinn	B
Ryan	B
Sophia	A
Tyler	F

1.1.6 Step 5: Analysis 1

Let's analyze the data so far. You do not need to run additional code here.

Answer the following questions in Markdown cells:

1. Who is the top-performing student based on average score?

2. Who is the lowest-performing student?
3. How many students received **Grade A**?
4. How many students received **Grade F**?
5. Based on questions 3 and 4, what can you tell?

1.1.7 Step 5 Answers

- 1) Fiona with an average of 95
- 2) George with an average of 64.33
- 3) 6 students; Charlie, Julia, Liam, Olivia, Sophia, Fiona
- 4) 2 students; George, Tyler
- 5) There are three times as many students with As than Fs, and the distribution of average scores bucketed by grade letters appears negatively skewed.

1.1.8 Step 6: Calculate Original Class Averages

Now, let's calculate the class averages. Compute **overall class averages** in each subject.

- Step 1: First, create three total variables: `math_total`, `science_total`, `english_total`
- Step 2: Use a loop and find the class average of each subject.
- Step 3: Store the results as: `original_math_avg`, `original_science_avg`, `original_english_avg`
- Step 4: Print the results.

```
[3]: #step 1
math_total = 0
science_total = 0
english_total = 0

#step 2
for student in students:
    math_total += student[1]
    science_total += student[2]
    english_total += student[3]

#step 3
original_math_avg = math_total/len(students)
original_science_avg = science_total/len(students)
original_english_avg = english_total/len(students)

#step 4 the fancy way using methods from class
ClassTotals = [{"Class", "Total", "Average"},
               ["Math", math_total, original_math_avg],
               ["Science", science_total, original_science_avg],
               ["English", english_total, original_english_avg]]
for row in ClassTotals:
    print(row[0], row[1], row[2], sep="\t")
```

```
Class    Total    Average
```

Math	1621	81.05
Science	1653	82.65
English	1652	82.6

1.1.9 Step 7: Analysis 2

What does it mean? Using your results from Step 6, explain your observation using the numerical averages in a Markdown cell.

1.1.10 Step 7 analysis answer

The students appear to be performing collectively in each class at approximately the same level, with slight variation found in math, which has an average score ~1.6 below science and English.

1.1.11 Step 8: Modify Dataset

Two students, Henry and Bianca, transferred into the class, and one student, Bob, moved out. Let's create a new dataset.

- Step 1: Add information about Henry and Bianca in alphabetical order.
 - ["Henry", 74, 88, 83]
 - ["Bianca", 62, 64, 60]
- Step 2: Remove Bob from the course list.

1.1.12 Step 9: Calculate New Class Averages

Have the class averages changed now? Recalculate class averages.

- Step 1: Reset total variables to 0.
- Step 2: Recalculate subject totals using a loop.
- Step 3: Find new_math_avg, new_science_avg, new_english_avg
- Step 4: Print all three new averages.

```
[4]: #step 8.1
new = [["Henry", 74, 88, 83], ["Bianca", 62, 64, 60]]
for student in new:
    total = sum(student[1:4])
    average = total / 3
    if average < 70:
        grade = "F"
    elif average < 80:
        grade = "C"
    elif average < 90:
        grade = "B"
    else:
        grade = "A"
    student.append(grade)
    students.append(student)
students.sort()
```

```

#step 8.2
for student in students:
    if student[0] == "Bob":
        students.remove(student)
        break

#step 9.1
math_total = 0
science_total = 0
english_total = 0

#step 9.2
for student in students:
    math_total += student[1]
    science_total += student[2]
    english_total += student[3]

#step 9.3
new_math_avg = math_total/len(students)
new_science_avg = science_total/len(students)
new_english_avg = english_total/len(students)

#step 9.4
ClassTotals = [["Class", "Total", "Average"],
               ["Math", math_total, new_math_avg],
               ["Science", science_total, new_science_avg],
               ["English", english_total, new_english_avg]]
for row in range(0, len(ClassTotals)):
    if row > 0:
        print(ClassTotals[row][0], ClassTotals[row][1], ␣
        ↪round(ClassTotals[row][2], 2), sep="\t")
    else:
        print(ClassTotals[row][0], ClassTotals[row][1], ClassTotals[row][2], ␣
        ↪sep="\t")

```

Class	Total	Average
Math	1679	79.95
Science	1723	82.05
English	1715	81.67

1.1.13 Step 10: Analysis 3

Using your results from the last two steps, analyze and explain the changes in each subject. What would be the cause of those changes?

1.1.14 Step 10 analysis answer

With the addition of Henry and Bianca the picture significantly changed. The class average for math has dropped ~1.1 points, looking at each student individually it becomes clear that both are struggling with math. This addition drops the math class average to a C with ~79.95. English also sees a slight drop of ~1 point and science also drops by ~0.6 points.

1.1.15 Step 11: Reflection

Write a short reflection (aim for at least 3 sentences in paragraphs) on your experience completing this assignment. You may write about what was challenging, mental breakthroughs, what you learned, etc.

1.1.16 Reflection

This project was interesting. The programming alone was fun, but it also highlighted how one data point could make a significant difference in your analysis and if you are calculating totals it's very important to include a recalculation of any summary statistics you may have. The most interesting part for me on the programming side was structuring the requested outputs because string formatting is always something in any language I try to avoid as much as possible. Especially the loop I used in step nine was interesting because it required an if statement because of the table header. I vaguely remember an easier way to tell python to treat the first row of a table as headers and the rest format as ints and doubles but it might be from a package or something.